

5.0 Summative Assessment: New Office Building Network Infrastructure Design Proposal

Executive Summary & Introduction

The new office building requires a secure and reliable switched network supporting HR, Finance, and IT. Each department will be assigned to a separate VLAN to provide logical segmentation, reduce unnecessary broadcast traffic, and help protect departmental resources.

IEEE 802.1Q trunk links will connect the switches and transport traffic for the departmental VLANs throughout the building. Rapid Spanning Tree Protocol (RSTP) will protect the Layer 2 network from switching loops, while redundant links and Link Aggregation Control Protocol (LACP) can improve availability and bandwidth. Secure switch-management practices will restrict administrative access.

Enterprise Network Architecture

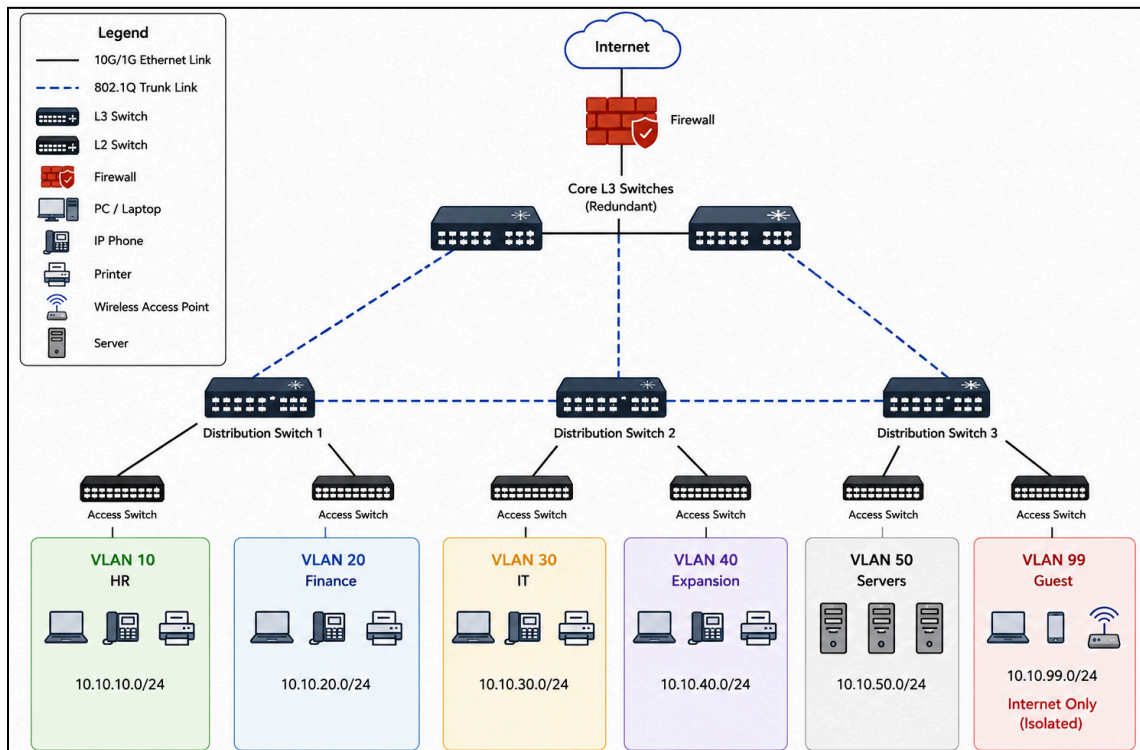


Figure 1. High-level network architecture illustrating redundant switching, VLAN-based departmental segmentation, shared network services, and secure cloud connectivity.

VLAN Segmentation & Trunking

The proposed network uses VLANs to logically separate HR, Finance, IT, servers, and guest devices while sharing the same physical switching infrastructure. Each VLAN creates an independent broadcast domain with its own IP subnet, reducing unnecessary traffic and improving performance, security, manageability, and troubleshooting.

IEEE 802.1Q trunking carries multiple VLANs between switches, allowing departmental VLANs to extend across the network. When communication between VLANs is required, Layer 3 switching provides inter-VLAN routing, while Access Control Lists (ACLs) control access according to business and security requirements.

Private IPv4 addressing and dual-stack IPv4/IPv6 provide capacity for current operations and future growth.

Department/ VLAN	IPv4 Network	Example IPv6 Prefix	Addressing	Enterprise Access
HR/ VLAN 10	10.10.10.0/24	2001:db8:100::/64	DHCP / IPv6 auto-configuration	DNS, AD, internal resources
Finance/ VLAN 20	10.10.20.0/24	2001:db8:200::/64	DHCP / IPv6 auto-configuration	DNS, AD, authorized resources
IT/ VLAN 30	10.10.30.0/24	2001:db8:300::/64	DHCP / IPv6 auto-configuration	DNS, AD, authorized resources
VLAN Expansion VLAN 40	10.10.40.0/24	2001:db8:400::/64	DHCP / IPv6 auto-configuration	DNS, AD, authorized resources
Server VLAN 50	10.10.50.0/24	2001:db8:500::/64	Static	Hosts centralized services
Guest VLAN 99	10.10.99.0/24	2001:db8:990::/64	DHCP / IPv6 auto-configuration	Internet only; isolated from internal LAN

*The IPv6 addresses shown use the IETF documentation prefix **2001:db8::/32** to illustrate the addressing plan. Production IPv6 prefixes would be based on addresses assigned to the organization.*

IPv4 Addressing & Subnetting

The network uses private IPv4 addressing with a separate /24 subnet for each VLAN. Each subnet supports up to 254 usable host addresses, providing sufficient capacity for current devices and future growth while maintaining a simple, consistent addressing structure.

Dept/Org	# Users	# of Devices	Network /Mask	Host Range	Broadcast
HR	40	100	10.10.10.0/24	10.10.10.1 - 10.10.10.254	10.10.10.255
Finance	50	125	10.10.20.0/24	10.10.20.1 - 10.10.20.254	10.10.20.255
IT	70	175	10.10.30.0/24	10.10.30.1 - 10.10.30.254	10.10.30.255
VLAN Expansion	100	250	10.10.40.0/24	10.10.40.1 - 10.10.40.254	10.10.40.255
Servers	Devices	30	10.10.50.0/24	10.10.50.1 - 10.10.50.254	10.10.50.255
Guest VLAN	100	200	10.10.99.0/24	10.10.99.1 - 10.10.99.254	10.10.99.255

Switching Technologies & Features

Managed Layer 2 access switches connect endpoints, while redundant Layer 3 switches provide backbone connectivity and inter-VLAN routing. The following technologies support availability, performance, and secure switch access.

Feature	Purpose
Rapid Spanning Tree Protocol (RSTP)	Prevents Layer 2 switching loops and provides rapid convergence
Link Aggregation Control Protocol (LACP)	Bundle multiple uplinks for additional bandwidth and redundancy
802.1X Network Access Control	Authenticate users/devices before network access
Port Security	Limit MAC addresses and restricts unauthorized devices on access ports
Bridge Protocol Data Unit (BPDU) Guard	Protects the spanning-tree topology from unauthorized switches

Feature	Purpose
SSH	Provides encrypted administrative switch access
Management VLAN	Separates switch-management traffic from user traffic

Cloud Concepts & Connectivity

A hybrid cloud model combines on-premises services with scalable cloud resources. Critical services such as DNS, DHCP, and authentication can remain local, while cloud services support applications, backup, disaster recovery, and remote accessibility. Connectivity options vary according to security, bandwidth, latency, scalability, and cost requirements.

Cloud Deployment & Infrastructure Concepts

Cloud Concept	Enterprise Use
Public Cloud	Microsoft 365, backups, collaboration, web services
Private Cloud	Dedicated applications and sensitive workloads
Hybrid Cloud	Combines on-premises infrastructure with cloud services
SaaS	Complete hosted applications
IaaS	Virtualized compute, storage, and networking resources
PaaS	Hosted application-development platform
Elasticity	Ability to dynamically increase or decrease resources
Scalability	Ability to increase capacity as requirements grow

Cloud Connectivity

Cloud connectivity securely links users, enterprise networks, and cloud resources. The appropriate method depends on requirements for security, bandwidth, latency, cost, and scalability.

Connectivity Option	Enterprise Use
Internet/SaaS	Access cloud-hosted business applications
Remote-Access VPN	Secure access for individual remote users
Site-to-Site VPN	Encrypted connection between organizational networks/cloud environments
Dedicated Cloud Connection	Higher-bandwidth, lower-latency private cloud connectivity

Networking Appliances, Applications & Functions

Component	Primary Function	How It Differs
Layer 2 Switch	Connects devices within VLANs	Forwards frames primarily by MAC address
Layer 3 Switch	Switching + inter-VLAN routing	Routes between departmental VLANs
Router	Connects different IP networks	Commonly connects LAN/WAN/external networks
Firewall	Filters and inspects traffic	Enforces security policy rather than simply forwarding traffic
Wireless AP	Provides WLAN connectivity	Bridges wireless clients into the wired network
IDS/IPS	Detects/blocks malicious traffic	Security monitoring/protection rather than general connectivity
DHCP	Automatically provides IP configuration	Address-configuration service
DNS	Resolves names to IP addresses	Name-resolution service

Security Considerations

Switches will be administered using SSH, with administrative access restricted to authorized IT personnel through a dedicated management VLAN. Port security, 802.1X, BPDU Guard, and ACLs provide additional protection against unauthorized devices, switching threats, and inappropriate inter-VLAN access.

Conclusion

The proposed network uses departmental VLANs and IEEE 802.1Q trunking to provide secure and efficient connectivity for HR, Finance, and IT. RSTP, LACP, secure switch management, and access controls improve availability and security, while cloud connectivity and supporting network services provide flexibility for current operations and future growth.

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